

北京邮电大学

《网络安全》课程实验报告

实验名称: 实验 5: 入侵检测实验

学生姓名: _____

学 号: _____

班 级: _____

实验 5：入侵检测课堂实验

一、实验目的

- 1.在 linux 平台建立基于 Snort 的 IDS。
- 2.在部署 snort 的本机建立规则，实现对 dos 攻击、nmap 扫描和 ssh 连接的检测。

二、实验环境

- 1.虚拟机：ubuntu、kali
- 2.工具：nmap、snort、hping3

三、实验过程与结果

3.1 步骤一：课前准备

1.安装 snort

```
sudo apt install -y snort
```

下载完成后，会提示进行相应配置，按要求配置即可

首先是输入 CIDR 格式的本地网段（例如我的主机 ip 是 192.168.204.136，所以配置为 192.168.0.0/16）。其次是填写网卡位置，网卡位置可通过 ifconfig 指令查看，此图中即为 ens33

```
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.204.136 netmask 255.255.255.0 broadcast 192.168.204.255
    inet6 fe80::64ee:296:8399:66eb prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:6d:cc:fd txqueuelen 1000 (Ethernet)
    RX packets 23927 bytes 29763885 (29.7 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8822 bytes 954646 (954.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2.配置一些必要文件夹

(1) Snort 的安装目录

```
sudo mkdir -p /etc/snort/rules/iplists
sudo mkdir -p /etc/snort/preproc_rules
sudo mkdir /usr/local/lib/snort_dynamicrules
sudo mkdir /etc/snort/so_rules
```

```
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo mkdir -p /etc/snort/rules/iplists
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo mkdir -p /etc/snort/preproc_rules
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo mkdir /usr/local/lib/snort_dynamicrules
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo mkdir /etc/snort/so_rules
```

(2) 存储过滤规则和服务器黑白名单

```
sudo touch /etc/snort/rules/iplists/default.blacklist
sudo touch /etc/snort/rules/iplists/default.whitelist
sudo touch /etc/snort/rules/so_rules
```

```
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo touch /etc/snort/rules/iplists/default.blacklist
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo touch /etc/snort/rules/iplists/default.whitelist
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo touch /etc/snort/rules/so_rules
```

(3) 创建日志目录

```
sudo mkdir /var/log/snort
sudo mkdir /var/log/snort/archived_logs
```

```
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo mkdir /var/log/snort
mkdir: cannot create directory '/var/log/snort': File exists
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo mkdir /var/log/snort/archived_logs
```

(4) 调整权限

```
sudo chmod -R 5775 /etc/snort
sudo chmod -R 5775 /var/log/snort
sudo chmod -R 5775 /var/log/snort/archived_logs
sudo chmod -R 5775 /etc/snort/rules/so_rules
sudo chmod -R 5775 /usr/local/lib/snort_dynamicrules
```

```
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo chmod -R 5775 /etc/snort
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo chmod -R 5775 /var/log/snort
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo chmod -R 5775 /var/log/snort/archived_logs
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo chmod -R 5775 /etc/snort/rules/so_rules
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo chmod -R 5775 /usr/local/lib/snort_dynamicrules
```

3.修改默认配置

```
sudo nano /etc/snort/snort.conf
```

(1) 首先根据自己的 ip 配置网络信息 (45 行左右), 我的主机 ip 是 192.168.204.136, 所以配置为 192.168.0.0/16

```
# Setup the network addresses you are protecting
# ipvar HOME_NET 192.168.0.0/16
```

然后修改默认文件路径 (111 行左右)

```
# Path to your rules files (this can be a relative path)
# Note for Windows users: You are advised to make this an absolute path,
# such as: c:\snort\rules
var RULE_PATH /etc/snort/rules
var SO_RULE_PATH /etc/snort/so_rules
var PREPROC_RULE_PATH /etc/snort/preproc_rules

# If you are using reputation preprocessor set these
# Currently there is a bug with relative paths, they are relative to where snort is
# not relative to snort.conf like the above variables
# This is completely inconsistent with how other vars work, BUG 89986
# Set the absolute path appropriately
var WHITE_LIST_PATH /etc/snort/rules/iplists
var BLACK_LIST_PATH /etc/snort/rules/iplists
```

(2) 在配置完成后, 可以通过 `sudo snort -c /etc/snort/snort.conf -A fast -l /var/log/snort -i ens33` 命令, 简单测试配置是否正确。若正确, 则将启动 snort

```
sudo snort -c /etc/snort/snort.conf -A fast -l /var/log/snort -i ens33
```

```
Decoding Ethernet

--== Initialization Complete ==--

o''~)~
''''

-*> Snort! <*-
Version 2.9.7.0 GRE (Build 149)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.9.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11

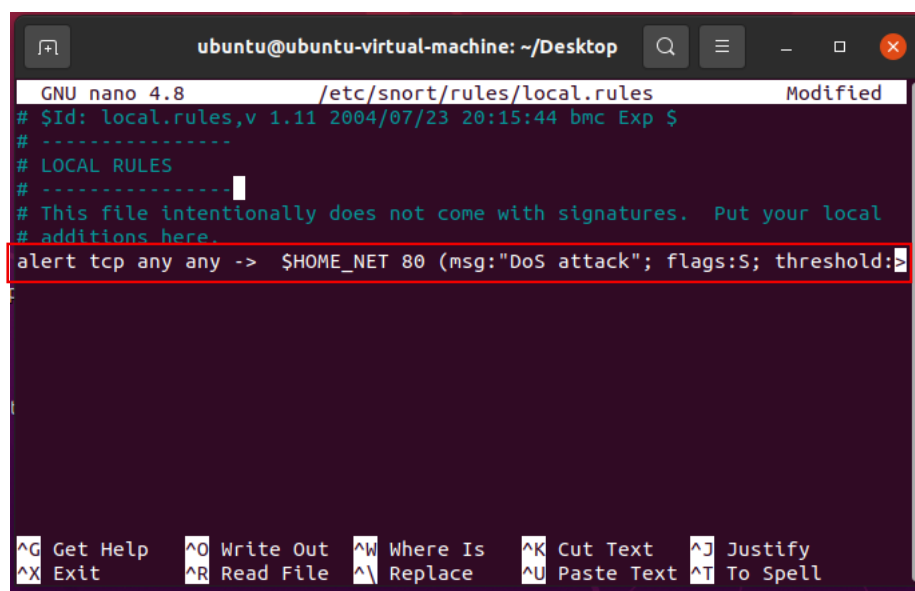
Rules Engine: SF_SNORT_DETECTION_ENGINE Version 2.4 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>

Commencing packet processing (pid=4551)
```

3.2 步骤二：Snort 检测 DoS（syn flood）攻击

1.Snort/rules 目录下的 local.rules 文件中，设置如下规则

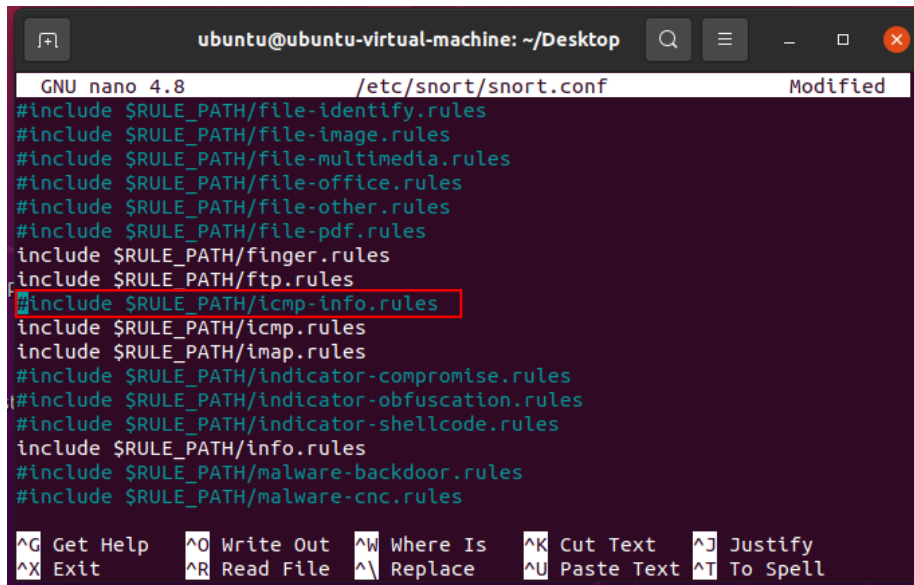
```
sudo nano /etc/snort/rules/local.rules
alert tcp any any -> $HOME_NET 80 (msg:"DoS attack"; flags:S; threshold:type
threshold, track by_dst, count 100, seconds 60; sid:2000003)
```



```
ubuntu@ubuntu-virtual-machine: ~/Desktop
GNU nano 4.8 /etc/snort/rules/local.rules Modified
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
alert tcp any any -> $HOME_NET 80 (msg:"DoS attack"; flags:S; threshold:
-----
^G Get Help      ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify
^X Exit          ^R Read File    ^L Replace      ^U Paste Text   ^T To Spell
```

2.打开 snort.conf，注释掉 include \$RULE_PATH/icmp-info.rules（608 行）

```
sudo nano /etc/snort/snort.conf
```

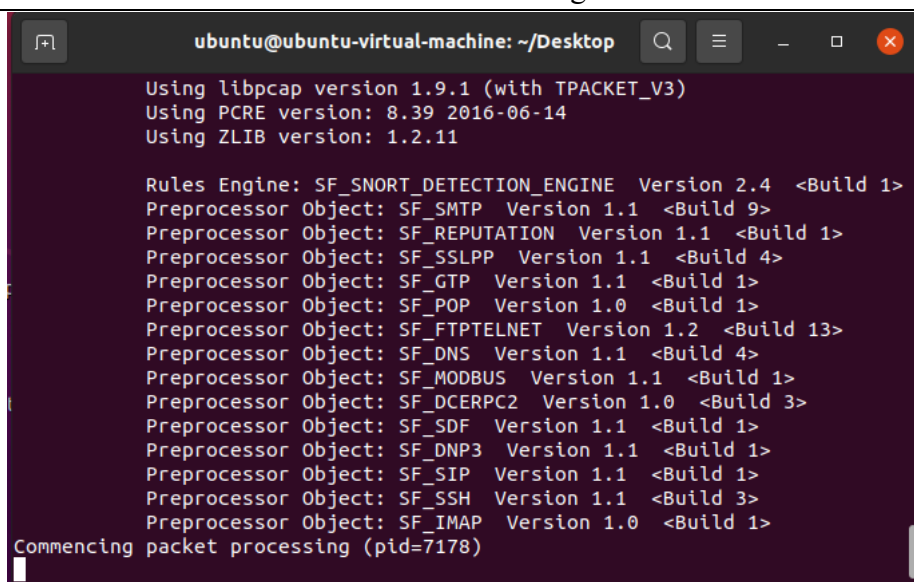


```
ubuntu@ubuntu-virtual-machine: ~/Desktop
GNU nano 4.8 /etc/snort/snort.conf Modified
#include $RULE_PATH/file-identify.rules
#include $RULE_PATH/file-image.rules
#include $RULE_PATH/file-multimedia.rules
#include $RULE_PATH/file-office.rules
#include $RULE_PATH/file-other.rules
#include $RULE_PATH/file-pdf.rules
include $RULE_PATH/finger.rules
include $RULE_PATH/ftp.rules
#include $RULE_PATH/icmp-info.rules
include $RULE_PATH/icmp.rules
include $RULE_PATH/imap.rules
#include $RULE_PATH/indicator-compromise.rules
#include $RULE_PATH/indicator-obfuscation.rules
#include $RULE_PATH/indicator-shellcode.rules
include $RULE_PATH/info.rules
#include $RULE_PATH/malware-backdoor.rules
#include $RULE_PATH/malware-cnc.rules

^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify
^X Exit      ^R Read File  ^\ Replace   ^U Paste Text ^T To Spell
```

3.启动 snort

```
sudo snort -c /etc/snort/snort.conf -A fast -l /var/log/snort -i ens33
```



```
ubuntu@ubuntu-virtual-machine: ~/Desktop

Using libpcap version 1.9.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11

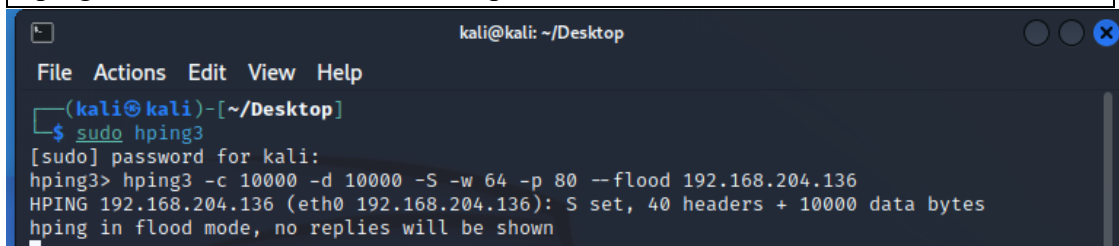
Rules Engine: SF_SNORT_DETECTION_ENGINE Version 2.4 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>

Commencing packet processing (pid=7178)
```

4.使用另一台 kali 使用 hping3 攻击（在 dos 实验中使用过的）。在日志中查看检测报告(cat /var/log/snort/alert)

```
sudo hping3
```

```
hping3 -c 10000 -d 10000 -S -w 64 -p 80 --flood 192.168.204.136
```



```
kali@kali: ~/Desktop

File Actions Edit View Help
(kali@kali)~[~/Desktop]
$ sudo hping3
[sudo] password for kali:
hping3> hping3 -c 10000 -d 10000 -S -w 64 -p 80 --flood 192.168.204.136
HPING 192.168.204.136 (eth0 192.168.204.136): S set, 40 headers + 10000 data bytes
hping in flood mode, no replies will be shown
```

5.可以看到检测报告。

```
sudo cat /var/log/snort/alert
```

```
ubuntu@ubuntu-virtual-machine: ~/Desktop
ubuntu@ubuntu-virtual-machine: ~/Desktop
4:10879 -> 192.168.204.136:80
11/27-13:32:59.416185 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:11239 -> 192.168.204.136:80
11/27-13:32:59.654966 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:11518 -> 192.168.204.136:80
11/27-13:33:00.062774 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:11956 -> 192.168.204.136:80
11/27-13:33:00.242261 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:12098 -> 192.168.204.136:80
11/27-13:33:00.446114 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:12218 -> 192.168.204.136:80
11/27-13:33:00.640068 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:12318 -> 192.168.204.136:80
11/27-13:33:00.834979 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:12418 -> 192.168.204.136:80
11/27-13:33:01.037432 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:12518 -> 192.168.204.136:80
11/27-13:33:01.270456 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:12618 -> 192.168.204.136:80
11/27-13:33:01.465226 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:12718 -> 192.168.204.136:80
11/27-13:33:01.661753 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:12818 -> 192.168.204.136:80
11/27-13:33:01.868509 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:12918 -> 192.168.204.136:80
11/27-13:33:02.050457 [**] [1:2000003:0] DoS attack [**] [Priority: 0] {TCP} 192.168.204.13
4:13018 -> 192.168.204.136:80
ubuntu@ubuntu-virtual-machine:~/Desktop$
```

3.3 步骤三：Snort 检测 nmap 扫描

1.以 nmap NULL 扫描为例：同理在 local.rules 中写下如下规则：

```
sudo nano /etc/snort/rules/local.rules
alert tcp $EXTERNAL_NET any -> $HOME_NET any (msg:"Nmap Null
PortScan!"; flags:0; threshold:type both,track by_dst,count 20,seconds 60;
sid:10000012; rev:1;)
```

```
GNU nano 4.8 /etc/snort/rules/local.rules Modified
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
alert tcp $EXTERNAL_NET any -> $HOME_NET any (msg:"Nmap Null PortScan!"; flags:0; threshold>
alert tcp any any -> $HOME_NET 80 (msg:"DoS attack"; flags:S; threshold:type threshold, tr>
```

2.nmap NULL 扫描是 nmap 秘密扫描的一种，相对比较隐蔽，通过向目标主机端口发送标志位 flags 为 0 的包进行扫描。使用 kali 中的 nmap 对主机进行 NULL 扫描。

```
nmap -sN 192.168.204.136
```

```
(kali@kali)-[~/Desktop]
$ nmap -sN 192.168.204.136
You requested a scan type which requires root privileges.
QUITTING!
```



```
(root@kali)-[/home/kali/Desktop]
# nmap -sN 192.168.204.136

Starting Nmap 7.92 ( https://nmap.org ) at 2025-11-27 13:39 CST
Nmap scan report for 192.168.204.136
Host is up (0.0042s latency).
All 1000 scanned ports on 192.168.204.136 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:6D:CC:FD (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds
```

3.查看 snort 报警信息。

```
sudo cat /var/log/snort/alert
```

```
11/27-13:37:09.018755  [**] [1:10000012:1] "Nmap Null PortScan!" [**] [Priority: 0] {TCP} 19
2.168.204.134:54854 -> 192.168.204.136:143
11/27-13:37:09.103629  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attemp
ted Information Leak] [Priority: 2] {TCP} 192.168.204.134:54854 -> 192.168.204.136:705
11/27-13:37:09.200535  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Inf
ormation Leak] [Priority: 2] {TCP} 192.168.204.134:54854 -> 192.168.204.136:161
ubuntu@ubuntu-virtual-machine:~/Desktop$
```

3.4 步骤四：Snort 检测 ssh 连接

1.首先将两台机器（可以是主机和虚拟机）的 ssh 服务开启。

(1) kali 开启 ssh 服务命令

```
sudo /etc/init.d/ssh start
```

```
kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ sudo /etc/init.d/ssh start
[sudo] password for kali:
Starting ssh (via systemctl): ssh.service.
```

(2) 先安装 ssh 服务，后启动

```
sudo apt install openssh-server -y
```

```
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssh-server is already the newest version (1:8.2p1-4ubuntu0.13).
0 upgraded, 0 newly installed, 0 to remove and 442 not upgraded.
1 not fully installed or removed.
After this operation, 0 B of additional disk space will be used.
Setting up snort (2.9.7.0-5build1) ...
Progress: [ 33%] [#####.....]
```

(3) 启动服务，查看服务状态

```
sudo systemctl enable --now ssh
```

```
sudo systemctl status ssh
```

```

ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/sy
stemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
ubuntu@ubuntu-virtual-machine:~/Desktop$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: ena
   Active: active (running) since Thu 2025-11-27 13:49:30 PST; 4min 48s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 9180 (sshd)
      Tasks: 1 (limit: 4577)
     Memory: 1.0M
    CGroup: /system.slice/ssh.service
            └─9180 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

Nov 27 13:49:30 ubuntu-virtual-machine systemd[1]: Starting OpenBSD Secure Shell
Nov 27 13:49:30 ubuntu-virtual-machine sshd[9180]: Server listening on 0.0.0.0 >
Nov 27 13:49:30 ubuntu-virtual-machine sshd[9180]: Server listening on :: port >
Nov 27 13:49:30 ubuntu-virtual-machine systemd[1]: Started OpenBSD Secure Shell
lines 1-15/15 (END)

```

2.自主编写规则 local.rules，检测 ssh 连接行为，从使用到的端口角度，并添加到 rules 中。

```
sudo nano /etc/snort/rules/local.rules
```

```
alert tcp any any -> $HOME_NET 22 (msg:"ssh connection"; sid:100000003)
```

```

ubuntu@ubuntu-virtual-machine: ~/Desktop
ubuntu@ubuntu-virtual-machine: ~/... x ubuntu@ubuntu-virtual-machine: ~/... x
GNU nano 4.8 /etc/snort/rules/local.rules Modified
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
alert tcp any any -> $HOME_NET 22 (msg:"ssh connection"; sid:100000003)
alert tcp $EXTERNAL_NET any -> $HOME_NET any (msg:"Nmap Null PortScan!"; flags:
alert tcp any any -> $HOME_NET 80 (msg:"DoS attack"; flags:S; threshold:type t

```

3.连接 ssh

```
ssh ubuntu@192.168.204.136
```

```

ubuntu@ubuntu-virtual-machine: ~
File Actions Edit View Help
(kali@kali)-[~]
$ ssh ubuntu@192.168.204.136
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
ubuntu@192.168.204.136's password:
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-50-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

452 updates can be applied immediately.
374 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

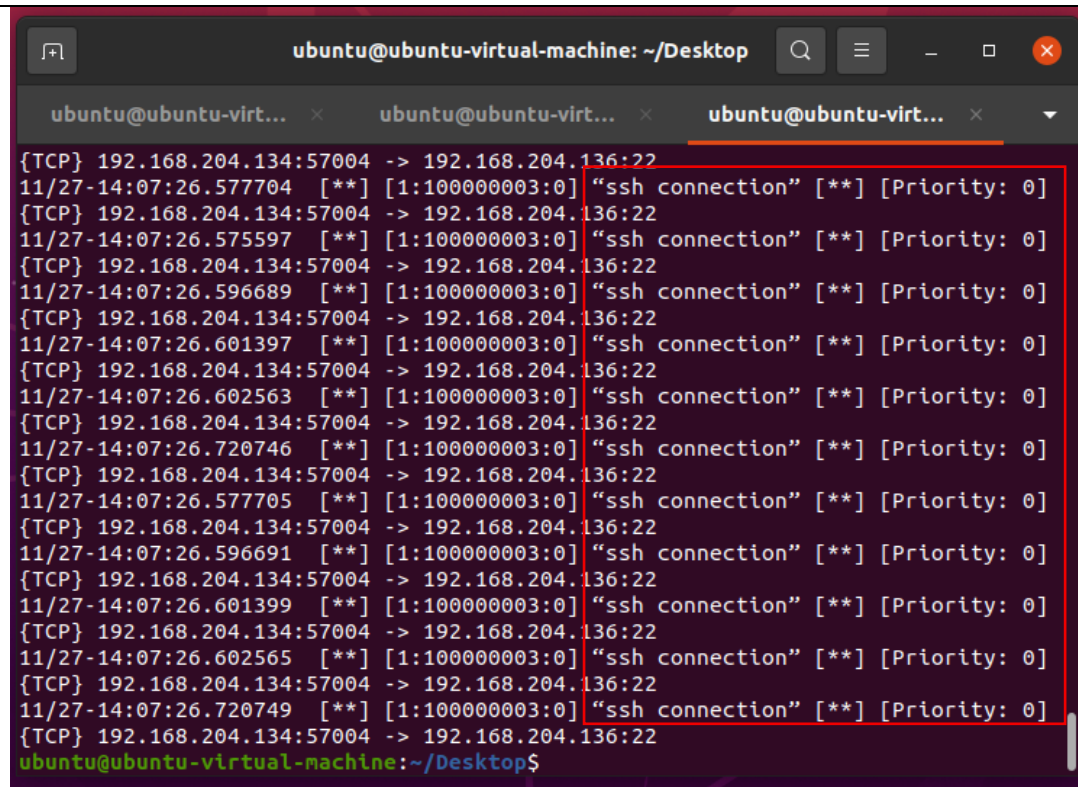
New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
Last login: Thu Nov 27 14:02:37 2025 from 192.168.204.134
ubuntu@ubuntu-virtual-machine:~$

```


4.查看 snort 信息，可以看到成功检测到了 ssh 连接！

```
sudo cat /var/log/snort/alert
```



The image shows a terminal window titled 'ubuntu@ubuntu-virtual-machine: ~/Desktop'. It displays the output of the command 'sudo cat /var/log/snort/alert'. The output consists of multiple lines of log entries, each representing an SSH connection detected by Snort. Each entry follows a similar pattern: a timestamp, a log level, a rule ID, a rule name, and a priority. The rule name is consistently 'ssh connection'. The priority is consistently 0. The log entries are separated by newlines. The terminal window has a dark background and a light-colored text. The title bar of the window is visible at the top, showing the window title and standard Ubuntu window controls (minimize, maximize, close).

```
ubuntu@ubuntu-virtual-machine: ~/Desktop
ubuntu@ubuntu-virt... x  ubuntu@ubuntu-virt... x  ubuntu@ubuntu-virt... x
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.577704  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.575597  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.596689  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.601397  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.602563  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.720746  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.577705  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.596691  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.601399  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.602565  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
11/27-14:07:26.720749  [**] [1:100000003:0] "ssh connection" [**] [Priority: 0]
{TCP} 192.168.204.134:57004 -> 192.168.204.136:22
ubuntu@ubuntu-virtual-machine:~/Desktop$
```

四、实验总结

在本实验中，我成功在 Linux 平台上建立了一套基于 Snort 的入侵检测系统（IDS）。整个实验的核心在于部署 Snort 引擎后，通过自主编写和配置检测规则，赋予其识别特定网络威胁的能力。我们针对三种典型的恶意或可疑行为——具有洪水特征的 DoS 攻击、用于信息搜集的 nmap 扫描以及潜在的未经授权 SSH 连接——精心设计了相应的检测规则。这些规则能够实时分析流经本机的网络流量，一旦匹配到预设的攻击模式，系统便会立即生成详细的警报日志，从而实现对目标威胁的有效监控与发现。

通过亲手完成从环境搭建、规则编写到功能测试的全过程，我们不仅掌握了 Snort 这一经典 IDS 工具的基本配置方法，更深入理解了如何根据具体攻击的特征（如异常流量、特殊标志位、特定端口与载荷）来构建检测逻辑。这次实践将理论知识与应用技能紧密结合，显著提升了对网络安全主动防御机制的认知水平和实操能力，为后续应对更复杂的安全挑战奠定了坚实的基础。